Gamma-ray Spectral Studies of B2 1811+31: Testing Theoretical Predictions for Emission in Relativistic Jets

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Background

What are we studying?

Active Galactic Nuclei are the bright central regions of galaxies whose central black holes are actively accreting mass.

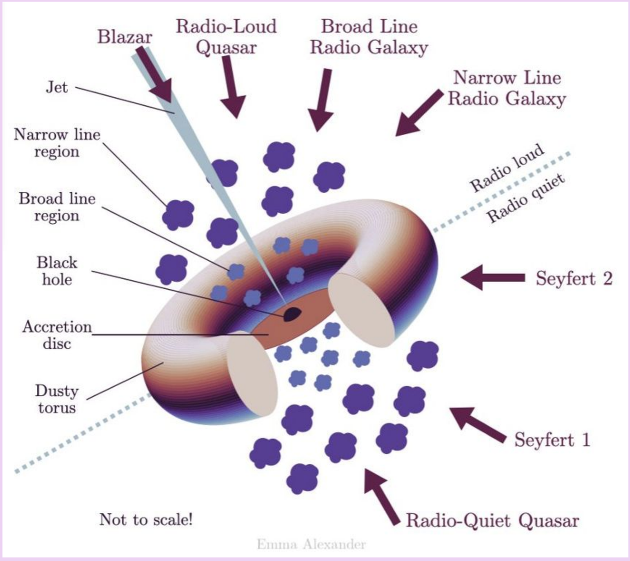

Figure 1. Artist's Scheme for AGN unification

Blazars are a type of active galactic nucleus that have relativistic jets pointed directly –or nearly so– at Earth. At the core, the supermassive black hole is actively accreting mass, which causes it to eject beams of accelerated matter from its polar regions at relativistic speeds. BL Lacs, a subclass of blazar, show rapid and large variabilities in their flux but the exact mechanisms that drive these variabilities are still unknown.

How can we understand the mechanisms driving blazar variabilities?

With gamma-ray observatories we can infer the accelerated particles in the blazars' jets.

VERITAS Gamma-Ray Observatory

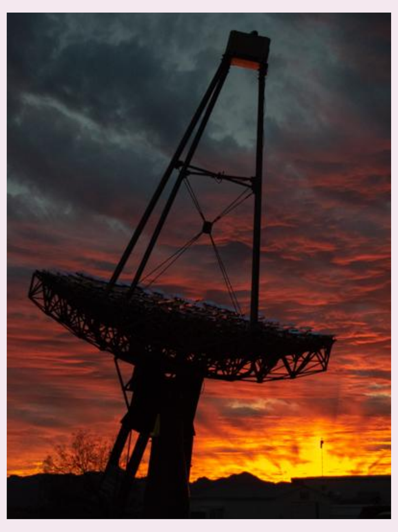

Figure 2a. T4 at sunset
credit: VERITAS picture of the day

- ★ Ground-based array of 4 gamma-ray telescopes
- ★ Indirectly observes gamma-ray events using Cherenkov light due to atmospheric interactions
- ★ Effective energy range ~ 85 GeV to 30 TeV

Fermi Large Area Telescope (Fermi-LAT)

- ★ Gamma-ray satellite
- ★ Unaffected by atmosphere
- ★ Effective range ~ 20 MeV to >300 GeV

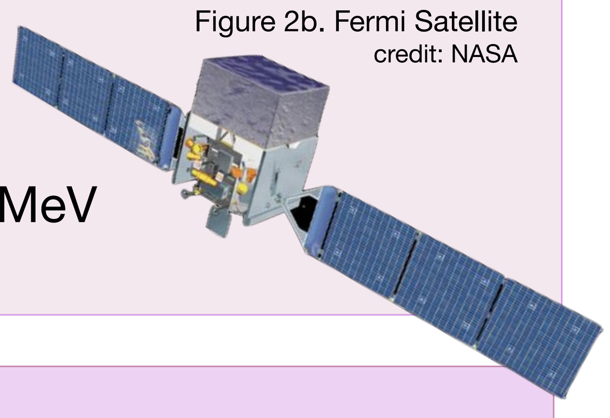

Figure 2b. Fermi Satellite
credit: NASA

Software

VEGAS

- ★ An analysis software package designed to analyze telescope data from VERITAS.
- ★ Soft cuts: allow us to cover a wide range of energies with the lowest energy threshold, which is important considering B2 1811+31 is an Intermediate BL Lac (IBL), which is a subclass known to peak between high and low energies.

EasyFermi & Fermipy

- ★ Software for carrying out analysis of gamma-ray data collected by the Fermi-LAT telescope.
- ★ easyFermi has a digital interface that aims to make the program accessible, allowing you to produce light curves, sed analyses, and statistical significance maps.

Gammapy

- ★ For gamma-ray analysis that allows one to produce joint fit plots using heterogeneous data and overplot data; helpful for comparing analyses like the Flare and Quiescent states.

Goals & Motivation

In October 2020, the Fermi-LAT recorded significant detections in the very high energy range coming from the location of B2 1811+31. While the source is significantly detected even in its quiescent state, showing a lower flux and a very flat light curve, this detection was interesting, as it showed a flaring flux.

In response, my goal is to create a spectrum for both flaring and quiescent states using VERITAS and Fermi data of the source and fit it to a spectral model.

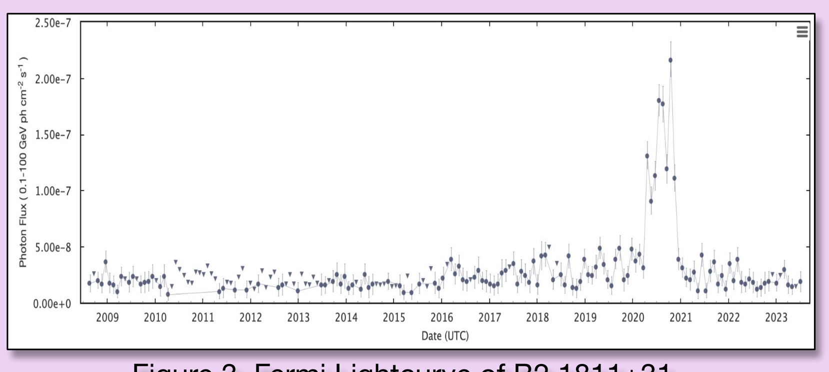

Figure 3. Fermi Lightcurve of B2 1811+31

Why?

- ★ Blazar spectra typically exhibit two characteristic peaks, one from synchrotron emission in the infrared to keV energies, and one from inverse compton emission at gamma-ray energies, where VERITAS and Fermi-LAT are sensitive
- ★ Developing spectral models of variable blazars will help us understand their spectral shapes and blazar variability.
- ★ Are the different flux states in blazars a result of a singular phenomenon, or can we see evidence in the flares of a new emission region?

Results & Discussion

In order to create a spectrum for both the flaring and quiescent states and fit it to any model, the first step was to acquire reasonable confirmation that the emissions were truly different. B2 1811+31 is an Intermediate BL Lac at redshift z = 0.117, meaning VERITAS has a good view within its energy threshold when the source is experiencing a flare.

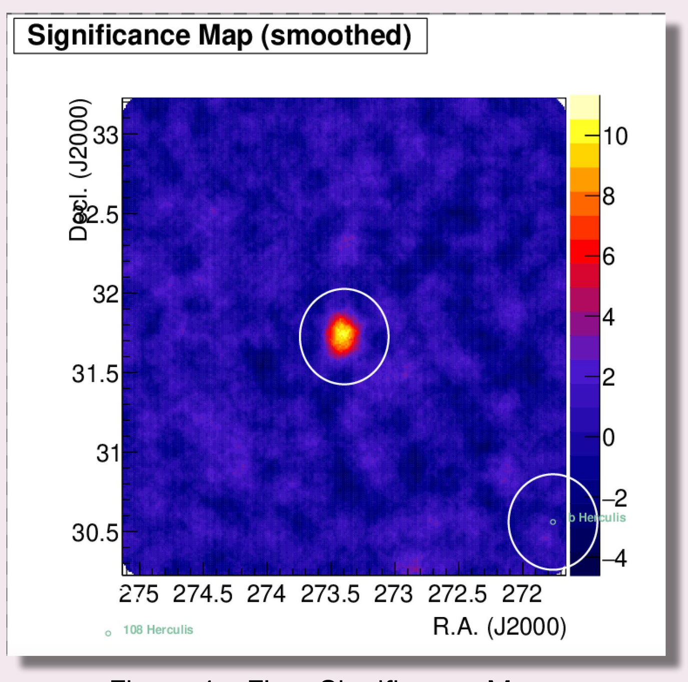

Figure 4a. Flare Significance Map

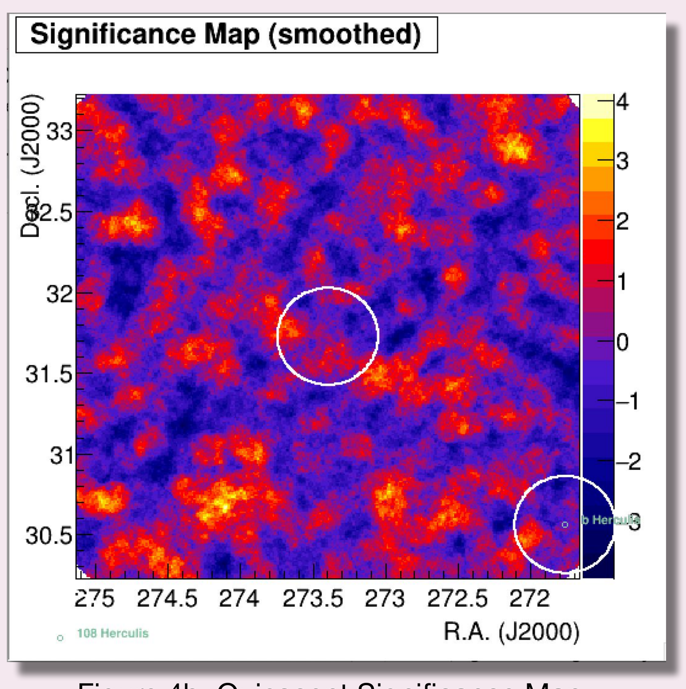

Figure 4b. Quiescent Significance Map

VEGAS generated significance maps for the VERITAS 4a. 9σ detection of B2 1811+31 flare from October 15, 2020 to October 19, 2020 to the left, and 4b. the region of B2 1811+31 recorded by VERITAS during a quiescent (low) stage of emission. It is important to note that the significance band (color scale to the left of each plot) is scaled to a maximum of 10σ in the case of the flare, and only 4σ in the quiescent, which is dominated by background.

Indeed, VERITAS was able to resolve a significant excess during the October 2020, whereas the data taken when the blazar was thought to not be flaring reports little statistical significance, and –as the following plots show– merely records upper limits, and means the emission is below that of the flaring state.

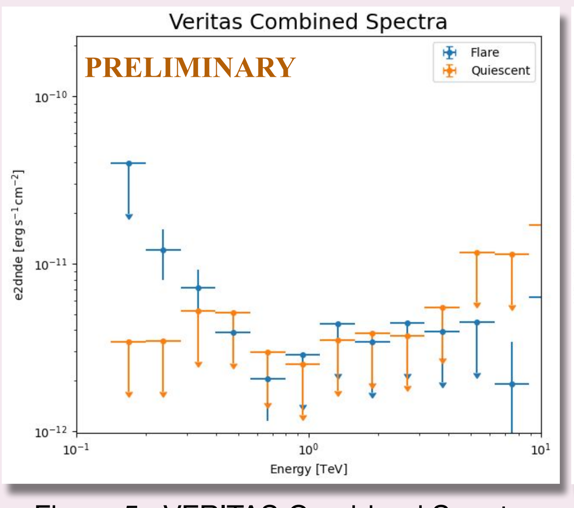

Figure 5a.VERITAS Combined Spectra

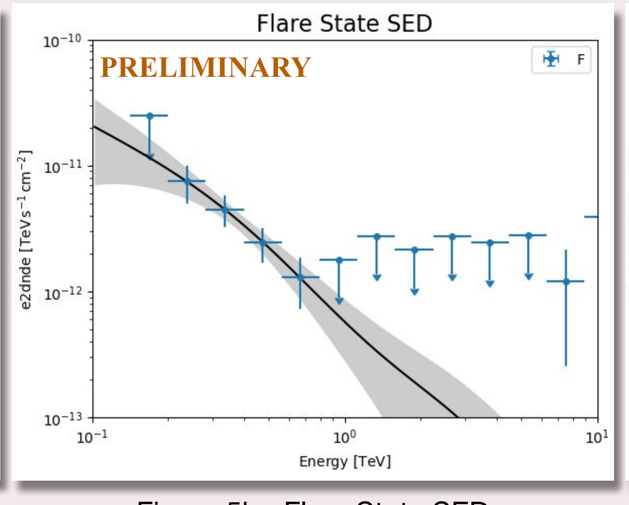

Figure 5b. Flare State SED

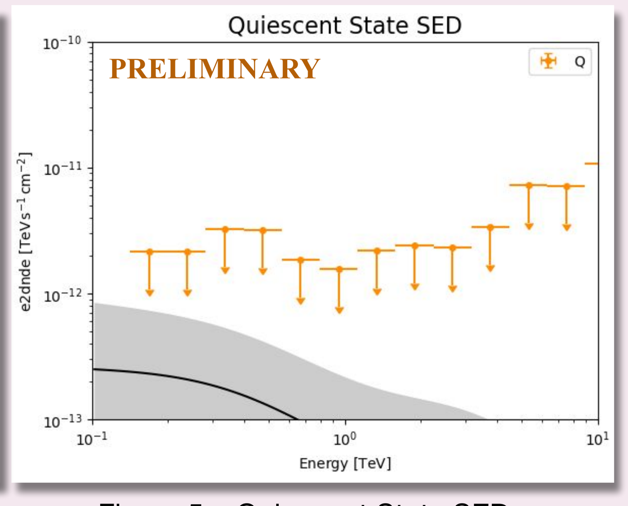

Figure 5c. Quiescent State SED

The above plots show the spectral energy distributions from data collected by VERITAS on both the flare and quiescent states, with upper limits for TS < 4. Figures 5b and 5c show the Flare and Quiescent data, respectively, fitted to an absorbed power law model.

- ★ Due to the soft spectrum of the IBL, the flux drops below VERITAS sensitivity above a TeV, but the combined spectrum supports the existence of a different spectral shape for each state, seeing as the flare has a larger, significant flux than the quiescent.
- ★ Significant detections have allowed for a reasonable fit in the flare SED.
- ★ The quiescent SED shows a projection of the shape of the quiescent SED using upper limits which can be improved with the addition of future multiwavelength data.

Takeaways & Next Steps

- ★ In its current state, we have set the groundwork for a the development of a spectra for the second peak of B2 1811+31, with the Fermi-LAT data available for both the flaring and quiescent periods.
- ★ A continuation of this analysis will expand the spectra into the lower energy ranges, as further additions from other observatories, perhaps even into the infrared, will help model a full two-hump spectrum for the source.
- ★ Studying the variabilities in blazar jets by modeling their spectra could help explain their underlying emission models.

References

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[2] Meyer, Manuel, et al. "Characterizing the Gamma-Ray Variability of the Brightest Flat Spectrum Radio Quasars Observed with the Fermi Lat." The Astrophysical Journal, vol. 877, no. 1, 2019, p. 39, doi:10.3847/1538-4357/ab1651.

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